

Software Requirements Specification (SRS)

Project Title: Prediction of IPL Match Outcome Using Machine Learning Techniques

Version: 1.0

Date: October 2025

1. Introduction

1.1 Purpose

The purpose of this project is to predict the outcome of Indian Premier League (IPL) cricket matches using machine learning algorithms. The system aims to help fans, analysts, and sponsors forecast match results based on several factors such as team performance, toss decisions, player statistics, and venue details.

1.2 Scope

This project develops a predictive system that uses supervised machine learning algorithms such as Random Forest, Logistic Regression, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) to determine the probable winner of an IPL match. It can be extended to include player performance prediction and man-of-the-match forecasting.

1.3 Definitions, Acronyms, and Abbreviations

IPL: Indian Premier League

ML: Machine Learning

SVM: Support Vector Machine

RFC: Random Forest Classifier

KNN: K-Nearest Neighbor

CSV: Comma Separated Values

1.4 References

Original paper: 'Prediction of IPL Match Outcome Using Machine Learning Techniques', Atlantis Highlights in Computer Sciences, 2021.

Kaggle IPL datasets (2008–2019).

2. Overall Description

2.1 Product Perspective

The system is a standalone analytical tool that reads historical IPL data, trains ML models, and predicts outcomes for future matches using Python-based libraries.

2.2 Product Features

- Data preprocessing and cleaning
- Feature extraction and selection
- Model training and testing
- Match outcome prediction
- Visualization of performance metrics

2.3 User Characteristics

Users include cricket analysts, journalists, data scientists, and sports enthusiasts with basic knowledge of cricket and statistics.

2.4 Constraints

- Dependent on quality and completeness of IPL datasets
- Requires at least 4GB RAM and Intel i3 or higher processor
- Needs Python 3.8+ environment

2.5 Assumptions and Dependencies

- Assumes datasets are accurate and publicly available
- Match conditions like weather or injuries are not considered
- Similar gameplay patterns continue over seasons

3. Specific Requirements

3.1 Functional Requirements

ID	Requirement Description	Priority
FR1	Allow users to upload IPL datasets.	High
FR2	Preprocess and clean datasets.	High
FR3	Train ML models using training data.	High
FR4	Test trained models on test data.	High
FR5	Display accuracy metrics for algorithms.	Medium
FR6	Predict outcome between two given teams.	High
FR7	Visualize data trends such as win percentage.	Medium
FR8	Export results and reports.	Low

3.2 Non-Functional Requirements

- Performance: System should process and generate results within 5 seconds for medium datasets.
- Scalability: Should support new season data without major modification.
- Accuracy: Random Forest expected to achieve around 88% accuracy.
- Usability: Interface should be user-friendly and visually informative.